

Erwin Quanten

Staff Android Engineer – Java, Kotlin, Android SDK, Jetpack Compose

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PROFESSIONAL SUMMARY

Staff Android Engineer with enterprise experience across logistics, digital commerce, industrial operations, fintech-style identity workflows, and SaaS platforms, building secure mobile products with **Kotlin 1.6–1.9**, **Java 8–17**, **Android SDK 23–35**, **Jetpack Compose 1.2–1.6**, **Room 2.4–2.6**, **Retrofit 2.x**, and **Hilt 2.4x–2.5x**. Specialized in modernizing legacy Java Android apps into Kotlin-first architectures, improving crash-free sessions, offline reliability, API synchronization, CI/CD release quality, accessibility, secure authentication, and measurable product performance while collaborating with backend, QA, DevOps, security, and product teams.

SKILLS

- **Mobile:** Android SDK 23–35, **Kotlin 1.6–1.9**, **Java 8–17**, Jetpack Compose 1.2–1.6, XML Views, Material Components 1.6–1.11, Navigation Component 2.5–2.7
- **Architecture:** MVVM, Clean Architecture, Repository Pattern, Modular Android, Multi-module Gradle, Dependency Injection, Offline-first design
- **Android Jetpack:** ViewModel, LiveData, Flow, Room 2.4–2.6, WorkManager 2.7–2.9, DataStore 1.x, Paging 3.x, CameraX 1.x
- **Networking:** Retrofit 2.x, OkHttp 3.x–4.x, REST APIs, GraphQL Android Client, WebSocket handling, OAuth2, JWT, Certificate Pinning
- **Dependency Injection:** Hilt 2.4x–2.5x, Dagger 2.x, Koin 3.x
- **Database & Storage:** Room, SQLite, EncryptedSharedPreferences, DataStore, SQLCipher, Firebase Realtime Database, Firestore
- **Testing:** JUnit 4–5, Mockito 4–5, MockK 1.x, Espresso 3.x, Robolectric 4.x, Firebase Test Lab, Play Console Pre-launch Reports
- **CI/CD & Release:** Gradle 7–8, Android Gradle Plugin 7–8, GitHub Actions, Bitrise, Jenkins, Fastlane, Firebase App Distribution, Play Console
- **Monitoring:** Firebase Crashlytics, Performance Monitoring, Sentry Android SDK, Logcat, ANR diagnostics, Play Console Vitals
- **Cloud & Backend Integration:** Firebase, AWS API Gateway, AWS Cognito, S3, Lambda-backed APIs, RESTful microservices, secure API integration

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

=Houston, TX

- Led the modernization of a logistics SaaS Android platform from legacy **Java 8** activities into **Kotlin 1.8**, **MVVM**, and modular Android architecture, improving feature delivery speed by **24%**.
- Designed secure mobile authentication flows using **OAuth2**, JWT refresh handling, biometric unlock, and encrypted local storage to support enterprise users with stricter compliance requirements.
- Migrated critical screens from XML-based layouts into **Jetpack Compose 1.4–1.6**, reducing UI regression defects by **18%** and making reusable design components easier for multiple squads.
- Resolved a high-impact offline synchronization bug where duplicate shipment status updates were created after poor network recovery by introducing idempotency keys and **WorkManager 2.8** retry policies.
- Implemented background job orchestration for document uploads, geolocation pings, and status confirmations using **Kotlin Coroutines**, Flow, and structured concurrency to reduce stuck tasks by **21%**.
- Improved API reliability by replacing inconsistent callback-based Retrofit usage with coroutine suspend functions, centralized error mapping, and standardized timeout handling across mobile modules.
- Created a reusable **Room 2.5** offline cache layer with conflict resolution, timestamp comparison, and sync queues, helping field teams continue workflows during low-connectivity warehouse operations.
- Reduced crash rate by **27%** after analyzing Firebase Crashlytics ANR traces, fixing main-thread database access, large bitmap rendering issues, and lifecycle leaks in dashboard screens.
- Built CI/CD pipelines with **GitHub Actions**, Gradle 8, Fastlane, Firebase App Distribution, and Play Console release tracks to make QA builds and production deployments more predictable.
- Introduced Espresso, JUnit, MockK, and Firebase Test Lab coverage for authentication, shipment scanning, and offline sync flows, increasing release confidence across Android and backend teams.

- Partnered with backend engineers to stabilize REST contract changes by adding API version headers, typed response models, and backward-compatible parsing for older installed mobile clients.
- Mentored Android developers on **Kotlin**, Compose state management, modular architecture, and production debugging, creating shared mobile standards adopted across several product squads.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Owned Android development for digital commerce and customer-service mobile products using **Kotlin 1.4–1.7**, **Java 11**, Android SDK 26–33, Retrofit 2.x, Room, and Firebase.
- Migrated a large Java-based shopping application into a Kotlin-first codebase by converting service, repository, and ViewModel layers gradually without blocking active release cycles.
- Implemented product search, order tracking, push notifications, loyalty wallet, and secure checkout flows, improving mobile conversion stability and reducing failed order sessions by **16%**.
- Fixed a production crash caused by nullable backend fields in promotional pricing responses by introducing sealed result classes, strict DTO mapping, and defensive UI rendering.
- Reduced cold-start time by **22%** through lazy initialization, dependency graph cleanup, optimized Firebase startup, and removal of heavy synchronous logic from the application class.
- Built **Hilt 2.4x** dependency injection across feature modules, replacing manual singleton patterns and improving testability for networking, analytics, storage, and payment integration layers.
- Created reusable Compose and XML hybrid components during the transition period, allowing newer screens to use modern UI while older checkout screens stayed stable.
- Improved release quality by integrating Fastlane, Bitrise pipelines, Firebase App Distribution, Crashlytics quality gates, and Play Console staged rollout monitoring.
- Solved inconsistent session expiration issues by aligning Android token refresh behavior with backend gateway rules and adding retry-safe authentication interceptors in OkHttp.
- Collaborated with product, QA, backend, and DevOps teams to define mobile API standards, analytics naming conventions, accessibility rules, and release-readiness checklists.

Tessengerlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Developed Android applications for industrial operations, workforce mobility, and asset inspection using **Java 8**, early **Kotlin 1.2–1.4**, Android SDK 23–30, Retrofit, SQLite, and Firebase.
- Introduced **MVVM** architecture with ViewModel, LiveData, repositories, and testable service layers to replace tightly coupled activity-based logic in internal mobile tools.
- Built barcode scanning, checklist execution, image attachment, and offline inspection modules for plant operators, reducing manual paper-based reporting time by **19%**.
- Solved a recurring image upload failure caused by large compressed photos timing out on unstable plant Wi-Fi by adding background upload queues and adaptive compression.
- Migrated older Gradle and support-library projects into AndroidX, Material Components, and newer Android Gradle Plugin versions without breaking installed internal devices.
- Improved crash-free sessions by **23%** after fixing lifecycle observer leaks, inconsistent permission handling, and SQLite cursor management issues reported through Crashlytics.
- Implemented secure local storage for inspection drafts, authentication tokens, and device metadata using encrypted preferences and controlled cleanup after logout.
- Added automated unit tests and Espresso smoke tests for inspection forms, required-field validation, barcode flows, and offline retry behavior before production release.
- Worked flexibly across Android, backend API contracts, QA validation, and release support to keep manufacturing teams operational during phased mobile adoption.

Tessengerlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built and maintained Android applications for internal enterprise workflows using **Java 7–8**, Android SDK 16–23, SQLite, REST APIs, XML layouts, and support libraries.
- Implemented mobile forms, approval workflows, PDF viewing, inventory lookup, and role-based screens that helped operational teams replace spreadsheet-driven field processes.
- Fixed a serious rotation-state bug where partially completed forms lost user input by introducing saved instance state, local draft persistence, and lifecycle-safe validation.

- Improved data-entry completion speed by **17%** through better form grouping, client-side validation, faster lookup caching, and simpler navigation between operational screens.
- Integrated REST API communication with Retrofit 1.x–2.x migration planning, OkHttp interceptors, error handling, and authentication token storage for secured internal services.
- Supported migration from Eclipse/Ant Android projects into **Android Studio** and Gradle-based builds, improving developer setup consistency and release packaging quality.
- Created reusable Java utility classes for date formatting, network checks, file handling, and validation to reduce duplicated code across multiple internal Android applications.
- Collaborated with business analysts, backend developers, and QA testers to clarify acceptance criteria, reproduce device-specific bugs, and deliver stable mobile releases.

Tessengerlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed early Android and Java mobile features for internal business users using **Java 6–7**, Android SDK 8–15, SQLite, XML layouts, and REST-style service calls.
- Supported inventory, reporting, and employee task modules by converting desktop-oriented workflows into simpler Android screens for warehouse and field usage.
- Resolved device compatibility issues across older Android phones by adjusting layouts, memory usage, permission handling, and API-level conditional logic.
- Improved form submission reliability by **14%** after adding local validation, retry prompts, basic offline draft saving, and clearer user-facing error messages.
- Assisted senior engineers with debugging, release packaging, API testing, and documentation while building strong fundamentals in Android lifecycle and Java application design.

EDUCATION

KU Leuven — Master’s Degree, Computer Science

Sep 2003 – May 2009